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RESEARCH

Tiny deep learning models for real-time and efficient embedded driver state detection

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Abstract

Driver distraction and inattention are major contributors to traffic accidents, making reliable real-time driver state detection essential for intelligent transportation systems. Although deep learning (DL) has significantly improved detection accuracy, conventional models remain computationally demanding and are often unsuitable for deployment on embedded or low-power hardware. To address these limitations, this study benchmarks seven lightweight architectures—Efficient-Tiny, ESPNetv2-Small, MCUNet, Micro-MobileNet, PhiNets, ShuffleNetLite, and SqueezeNetv11—within a unified evaluation framework tailored for embedded deployment. Experimental results demonstrate that ESPNetv2-Small and SqueezeNetv11 achieved the highest accuracies of 99.50% and 95.81%, with compact model sizes of 0.74 MB and 0.33 MB and inference speeds of 40.49 FPS and 28.60 FPS, respectively. Efficient-Tiny, despite its reduced footprint (0.43 MB), reached 91.64% accuracy. These findings confirm that modern TinyDL architectures can deliver near state-of-the-art performance with minimal computational cost, offering a viable path toward energy-efficient, real-time driver monitoring in next-generation intelligent vehicles.

Keywords Deep learning · Lightweight models · Driver distraction detection · Real-time · Computer vision · TinyML · Embedded devices

1 Introduction

Deep learning (DL) [1, 2] has become foundational to modern computer vision, driving significant advancements in object detection, scene understanding, and human behavior analysis [3]. Convolutional neural networks (CNNs) and hybrid transformer-based architectures have achieved state-of-the-art results across multiple domains, including medical imaging, facial recognition, autonomous driving, and driver monitoring [4–6, 8]. However, deploying such models on embedded or low-power hardware remains challenging due to their substantial computational demands, memory footprint, and energy consumption [9].

To address these constraints, lightweight architectures, such as MobileNet [11], ShuffleNet [12], and EfficientNet variants [10], have been introduced. These models leverage techniques, such as depthwise separable convolutions, neural architecture search (NAS) [13], and compound scaling to reduce model complexity. Nevertheless, many existing approaches still rely on high-performance GPUs or CPUs and often neglect critical factors, such as convergence stability, activation memory usage, and inference latency—factors that are essential for reliable, real-time deployment in embedded systems.

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To bridge this gap, the present study investigates the performance, efficiency, and deployment feasibility of recent lightweight deep learning architectures for real-time driver state detection under constrained hardware conditions [14]. The goal is to identify models that balance predictive accuracy with computational and memory efficiency, enabling real-time inference on embedded platforms, such as Arduino, STM32, and ESP32 devices. The novelty of this work lies in its systematic benchmarking of seven state-of-the-art TinyDL architectures within a unified experimental framework, assessing not only accuracy but also training stability, test performance, model size, and inference speed. This comprehensive evaluation extends beyond conventional metrics by incorporating practical deployment considerations relevant to TinyML and embedded intelligence applications.

The main contributions of this research can be summarized as follows:

- A unified benchmarking framework is proposed to evaluate seven recent lightweight CNN architectures—Efficient-Tiny, ESPNetv2-Small, MCUNet, Micro-MobileNet, PhiNets, ShuffleNetLite, and SqueezeNetv11—specifically for real-time driver state detection tasks under embedded hardware constraints.
- A comparative analysis is conducted across training, validation, and test phases, focusing on convergence dynamics, generalization performance, and computational efficiency.
- Comprehensive hardware-oriented metrics are presented, including model size, RAM utilization, and inference speed, to assess deployability on low-power embedded platforms.
- Practical implications and optimization challenges are discussed to guide future TinyDL deployment efforts, including quantization, pruning, and multimodal expansion.

The remainder of this paper is organized as follows. Section 2 reviews related lightweight and real-time driver state detection models. Section 3 outlines the datasets, benchmark architectures, and evaluation methods. Section 4 presents and analyzes the experimental results. Sections 5 discuss deployment challenges and future research directions. Finally, Sect. 6 concludes the study.

2 Lightweight models for driver distraction detection

Lightweight real-time driver distraction detection has garnered increasing attention, as deep learning models must balance accuracy and computational efficiency to support deployment on embedded and resource-constrained platforms [6, 7]. Multiple research directions have emerged to address these constraints, including compact CNN architectures, optimized You Only Live Once (YOLO) variants, knowledge distillation frameworks, and hybrid CNN–Transformer models.

One prominent line of research focuses on knowledge distillation. Li et al. [4] introduced the Instance-Specific Multi-Teacher Knowledge Distillation (IsMt-KD) framework, which enhanced MobileNetV3-Small and ShuffleNetV2−0.5 by up to +8.41% on the StateFarm dataset, while sustaining real-time performance (42 FPS on MediaPad C5 and 25 FPS on Jetson TX2). Building on this, Nguyen et al. [15] proposed a lightweight Convolutional Neural Network (CNN) architecture with adaptive connections and a Convolutional Block Attention Module (CBAM), achieving 99.95% accuracy on the State Farm Distracted Driver Detection dataset (StateFarm) and 99.61% on the American University in Cairo version 2 dataset (AUC v2) with only 0.43M parameters, while maintaining high inference speeds across GPU, CPU, and NVIDIA Jetson Nano platforms. More recently, Liu et al. [19] developed a distillation-based Neural Architecture Search (NAS) and knowledge-transfer framework; their student model achieved 99.86–99.91% accuracy on StateFarm and 95.64% on the American University in Cairo Distracted Driver dataset version 2 (AUCD2) with just 0.42M parameters. A three-dimensional (3D) extension of their model further improved Drive&Act accuracy by up to 30%, while remaining compact (2.03M parameters).

A parallel research trajectory focuses on YOLO-based lightweight detectors. Lou and Nie [5] improved YOLOv5s by integrating GhostNet, Ghost Shuffle Convolution (GSConv), Convolutional Block Attention Module (CBAM), and Soft Non-Maximum Suppression (Soft-NMS), yielding a +1.5% increase in mean Average Precision (mAP) and a 7.6 giga–floating point operations (GFLOPs) reduction in computational cost, achieving 9 ms inference per image. Du et al. [18] extended YOLOv8n with Ghost Convolution (GhostConv), Ghost Cross-Stage Partial with Fusion (GhostC2f), Bidirectional Feature Pyramid Network (BiFPN), and Simple Attention Module (SimAM), reaching 96.3% mAP on the SFDDD with 39.3% fewer parameters and real-time inference at 75.76 frames per second (FPS) on NVIDIA Jetson TX2. Further improvements were introduced by Shen et al. [23], who proposed a Cross-Stage Partial–Star block (C2f–Star) and a Lightweight Shared Convolutional Detection (LSCD) head, attaining 99.6% accuracy on StateFarm while reducing model parameters and FLOPs by 56.4% and 45.1%, respectively.

Compact CNN models and TinyML-oriented designs continue to be widely explored. Huang et al. [16] introduced the Progressive Scalable Detection Network (PSD-Net), a pruned CNN model that achieved 96.4% mAP on

the Large-Scale Driver Behavior Dataset (LDDB) with 1.33M parameters and 52 FPS on Jetson TX2. Alajlan and Ibrahim [17] proposed the Drowsiness and Distraction Detection TinyML model, incorporating quantized variants of MobileNet and SqueezeNet to achieve up to 99.64% accuracy with model sizes as small as 0.05 MB. Khalil et al. [27] demonstrated a low-cost deployment scenario using a $15 Raspberry Pi Zero 2 W, running YOLOv8n enhanced with GhostNet and a head-pose estimation module, surpassing 90% mAP@50 at 10 FPS. Other contributions include Lv et al. [28], who integrated the SimAM and Coordinate Attention (CA) into MobileNet, reaching 99.80% accuracy on StateFarm with 3.17M parameters and 82.95 FPS. Gu et al. [26] developed the Adaptive Tiny Driver Detection Network (ATD2Net) for tiny-target detection, reporting 98.2% mAP on LDDB and 96.4% on StateFarm.

Hybrid CNN–Transformer architectures have also advanced the field. Tang et al. [21] proposed the Convolution and Transformer Network (CaTNet), combining a pruned ConvNeXt backbone with Transformer modules, which achieved 99.91% accuracy on the State Farm Distracted Driver Detection version 3 dataset (SFD3) with 2.832M parameters. Li et al. [22] introduced the Convolutional Vision Transformer (CoViT), integrating Multi-Scale Dilated Convolutions (MSDC), Depthwise Separable Convolutions (DSC), and a Lightweight Convolutional Attention Module (LCAM), achieving 95.17–97.89% accuracy across multiple datasets with an inference speed of 159 FPS. Semi-supervised approaches have also been explored. Mohammed et al. [20] developed a Vision Transformer (ViT)-based framework using pseudo-labeling and a teacher–student structure, achieving 95.43% on StateFarm and 70.39% on the Multimodal Multiview Driver Attention Dataset (3MDAD). Guo et al. [24] proposed the Lightweight Three-Level Attention network (L-TLA), which combines a dual-stream backbone with multi-level attention and contrastive learning, attaining 80.74% accuracy on Front Depth and 88.49% on the Multi-Angle Surveillance (MAS) dataset. Sun and Zhang [25] reported the Coordinate Attention–enhanced Convolutional Conditionally Parameterized Network (CCNET), an NAS-optimized, attention-enhanced CNN that achieved 99.8–99.9% accuracy on StateFarm with only 1.7M parameters and real-time performance of 54 FPS on Jetson Nano.

Despite these notable advances, many existing models—though lightweight—remain suboptimal for deployment in highly constrained TinyDL environments, such as microcontrollers and ultra-low-power edge hardware. Achieving a balanced trade-off between inference speed, energy efficiency, and generalization across diverse real-world driving scenarios remains a key open challenge, motivating further exploration in this study.

3 Methodology

This section outlines the methodological framework used to evaluate lightweight deep learning architectures for real-time driver state detection. The experimental design emphasizes fairness, reproducibility, and hardware relevance, and consists of three main components: dataset preparation, benchmark model selection, and standardized training configuration.

3.1 Datasets

The experiments in this study are primarily based on the State Farm Distracted Driver Detection dataset (SFDDD) [29], a widely adopted and comprehensive benchmark for driver distraction recognition. Compared to AUCD2 [30] and 100-Driver [31], SFDDD demonstrated the most consistent and reliable performance across the evaluated models.

The dataset comprises 22,424 labeled and 79,726 unlabeled RGB images with a resolution of 640×480 pixels, categorized into ten driver behavior classes including texting, phone usage, operating the radio, and interacting with passengers (see Fig. 1). All images were resized to 224×224, normalized to the [0, 1] range, and partitioned *driver-wise* into 70/15/15% splits for training, validation, and testing. This configuration was selected following the evaluation of multiple alternative data-splitting strategies, as it consistently demonstrated the most stable training dynamics and highest validation performance, while effectively preventing identity leakage across subsets.

To enhance robustness and generalization, a comprehensive data augmentation pipeline was applied. This included horizontal flipping, random rotation ($\pm 15°$), cropping, brightness and contrast adjustments, and random occlusion.

Overall, SFDDD provides a realistic, balanced, and challenging benchmark for assessing the performance of lightweight and TinyML-compatible architectures in embedded driver monitoring applications.

3.2 Benchmark architectures

This study compares seven state-of-the-art lightweight deep learning architectures—ShuffleNetV2, SqueezeNet, EfficientNet-Tiny, ESPNetv2, Micro-MobileNet, PhiNets, and MCUNet—designed for efficient real-time driver distraction detection.

3.2.1 ShuffleNetV2

ShuffleNetV2 is an optimized convolutional neural network that improves upon the original ShuffleNet by introducing

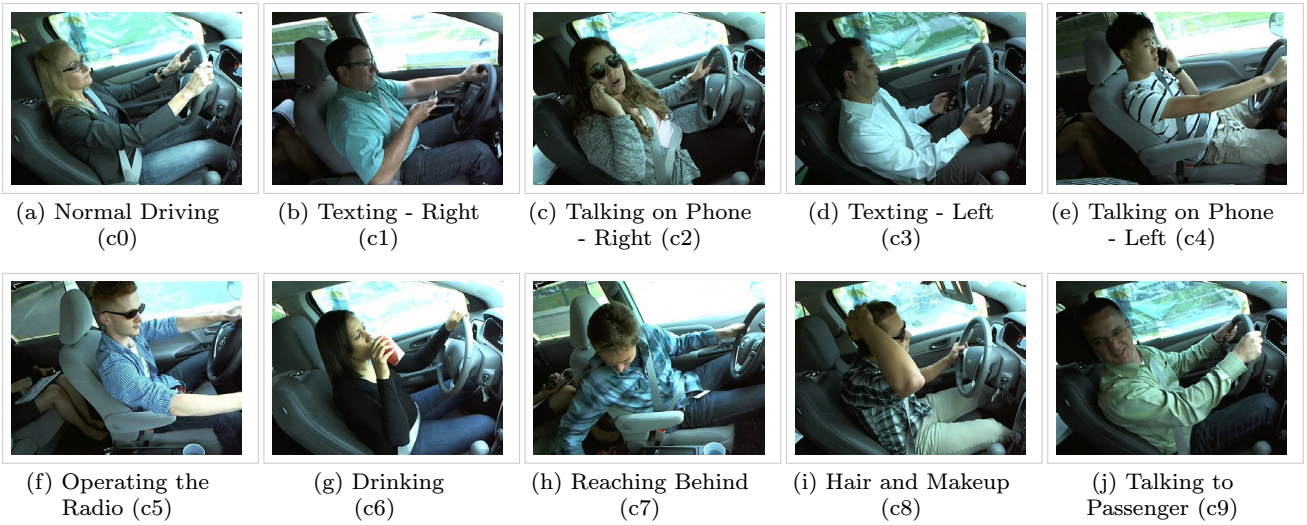


Fig. 1 Representative samples from the SFDDD dataset

a more balanced trade-off between computation cost and accuracy [32]. The key architectural innovation is the channel shuffle operation, which facilitates information exchange between grouped convolutions, thus reducing redundancy while preserving representational power. Each block consists of pointwise group convolutions, depthwise convolutions, and channel shuffling operations. Given an input tensor $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, the grouped convolution can be expressed as

$$\mathbf{Y} = \text{Concat}([\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_g]), \tag{1}$$

where g denotes the number of channel groups. The channel shuffle operation ensures inter-group communication by permuting feature maps across groups. ShuffleNetV2 achieves high computational efficiency, making it suitable for mobile and embedded devices.

3.2.2 SqueezeNet

SqueezeNet achieves AlexNet-level accuracy with approximately 50× fewer parameters through its compact fire module [33]. Each fire module consists of a squeeze layer using 1×1 convolutions followed by an expand layer composed of a mix of 1×1 and 3×3 filters. The squeeze operation reduces input channels, while the expand operation restores spatial complexity. Mathematically, for an input tensor $\mathbf{X}$

$$\mathbf{Y} = f_{3\times 3}(\sigma(f_{1\times 1}(\mathbf{X}))), \tag{2}$$

where σ is the activation function, and $f_{k\times k}$ denotes a convolution operation with kernel size $k\times k$. This design drastically reduces the number of parameters and FLOPs, enabling real-time execution on low-power processors.

3.2.3 EfficientNet-Tiny

EfficientNet-Tiny is derived from the EfficientNet family [13], which uses a compound scaling method to uniformly scale depth (d), width (w), and resolution (r) with a constant scaling coefficient ϕ. The network scaling is expressed as

$$d = \alpha^{\phi}, \quad w = \beta^{\phi}, \quad r = \gamma^{\phi}, \tag{3}$$

subject to the constraint $\alpha\cdot\beta^2\cdot\gamma^2 \approx 2$. This approach maintains a balance between accuracy and efficiency. The architecture uses inverted residual blocks with depthwise separable convolutions and squeeze-and-excitation (SE) modules, reducing memory footprint while preserving accuracy, which makes it ideal for TinyML deployment.

3.2.4 ESPNetv2

ESPNetv2 is designed for low-latency applications using an efficient spatial pyramid of dilated convolutions to capture multi-scale context [34]. The Efficient Spatial Pyramid (ESP) module splits the input feature maps into K groups, each processed with different dilation rates to expand the receptive field without increasing computational cost. Formally, the operation can be defined as

$$\mathbf{Y} = \sum_{k=1}^{K} f_{3\times 3}^{(r_k)}(\mathbf{X}), \tag{4}$$

where $f_{3\times 3}^{(r_k)}$ denotes a 3×3 convolution with dilation rate r_k. Using grouped pointwise convolutions followed by

depthwise dilated convolutions, ESPNetv2 achieves a favorable accuracy–latency trade-off for embedded vision systems.

3.2.5 Micro-MobileNet

Micro-MobileNet is a miniaturized version of MobileNetV1/V2 [12], designed explicitly for microcontrollers and ultra-low-power hardware. It relies on depthwise separable convolutions, which decouple spatial and channel-wise computations to reduce complexity. For an input tensor $\mathbf{X}$, the standard convolution operation is replaced by

$$\mathbf{Y} = f_{1\times1}(f_{DWC}(\mathbf{X})), \tag{5}$$

where f_{DWC} represents a depthwise convolution and $f_{1\times1}$ a pointwise convolution. This decomposition decreases the computational cost from $O(K^2C_{in}C_{out})$ to $O(K^2C_{in} + C_{in}C_{out})$. Micro-MobileNet variants (with width multipliers $\alpha \leq 0.5$) can be quantized to 8-bit precision for TinyML compatibility.

3.2.6 PhiNets

PhiNets are a family of networks optimized for energy efficiency on microcontrollers, focusing on the decoupling of memory, compute, and parameter constraints [35]. The architecture employs mixed-kernel convolutions and a hierarchical scaling function to minimize on-device memory usage. The optimization objective is formulated as

$$\mathcal{L} = \lambda_1 \cdot \text{MSE} + \lambda_2 \cdot \text{Params} + \lambda_3 \cdot \text{FLOPs}, \tag{6}$$

where λ_i are task-specific weights balancing accuracy and efficiency. PhiNets achieve up to 90% reduction in parameters compared to MobileNetV2 while maintaining competitive performance for image classification tasks.

3.2.7 MCUNet

MCUNet integrates neural architecture search (NAS) with a specialized inference engine (TinyEngine) to co-design both the model architecture and runtime execution for microcontrollers [36]. The framework optimizes for SRAM, Flash memory, and inference latency. The NAS optimization objective is expressed as

$$\min_{\theta} \ \mathcal{L}_{\text{task}}(\theta) + \lambda \cdot \mathcal{C}_{\text{latency}}(\theta), \tag{7}$$

where $\mathcal{L}_{\text{task}}$ represents the task loss and $\mathcal{C}_{\text{latency}}$ penalizes high latency configurations. MCUNet's co-design allows it to achieve ImageNet-level accuracy on devices with less than 1 MB Flash and 256 KB SRAM, making it one of the most efficient architectures for real-time edge deployment.

3.3 Training protocol

All models were trained under identical experimental conditions to ensure a fair and consistent comparison across architectures. A unified training workflow (Algorithm 1) was applied to all benchmark models—ShuffleNetV2, SqueezeNet, EfficientNet-Tiny, ESPNetv2, Micro-MobileNet, PhiNets, and MCUNet—thereby ensuring that any observed performance differences are attributable to architectural design rather than training variability.

Algorithm 1 Generalized training procedure for lightweight architectures

Input: Training dataset $\mathcal{D}_{train}$, validation dataset $\mathcal{D}_{val}$, model $\mathcal{M}_\theta$, epochs $E = 200$, batch size $B = 32$, learning rate $\eta = 10^{-3}$

Initialize: Model parameters θ using He or Xavier initialization

for epoch $e = 1$ to E **do**

- Shuffle and batch $\mathcal{D}_{train}$ into mini-batches of size B

  for each mini-batch (x, y) in $\mathcal{D}_{train}$ **do**

  - Normalize input: $\tilde{x} = x/255.0$
  - Forward pass: $\hat{y} = \mathcal{M}_\theta(\tilde{x})$
  - Compute loss using categorical cross-entropy:

$$\mathcal{L} = -\frac{1}{B}\sum_{i=1}^{B}\sum_{c=1}^{C} y_{i,c}\log(\hat{y}_{i,c}), \tag{8}$$

    where C is the number of classes, $y_{i,c}$ is the true label, and $\hat{y}_{i,c}$ the predicted probability.

  - Backpropagate gradients $\nabla_\theta\mathcal{L}$ and update θ using the Adam optimizer

  end for

- Evaluate $\mathcal{M}_\theta$ on $\mathcal{D}_{val}$ and record validation accuracy
- Reduce learning rate η if validation loss stagnates

end for

Return: Optimized model $\mathcal{M}_{\theta^*}$ with trained parameters

Each model was trained for 200 epochs using the Adam optimizer with an initial learning rate of 1×10^{-3} and categorical cross-entropy as the loss function. A dynamic learning rate scheduler was employed to reduce the learning rate after five consecutive epochs without improvement on the validation set. Training was conducted with a batch size of 32.

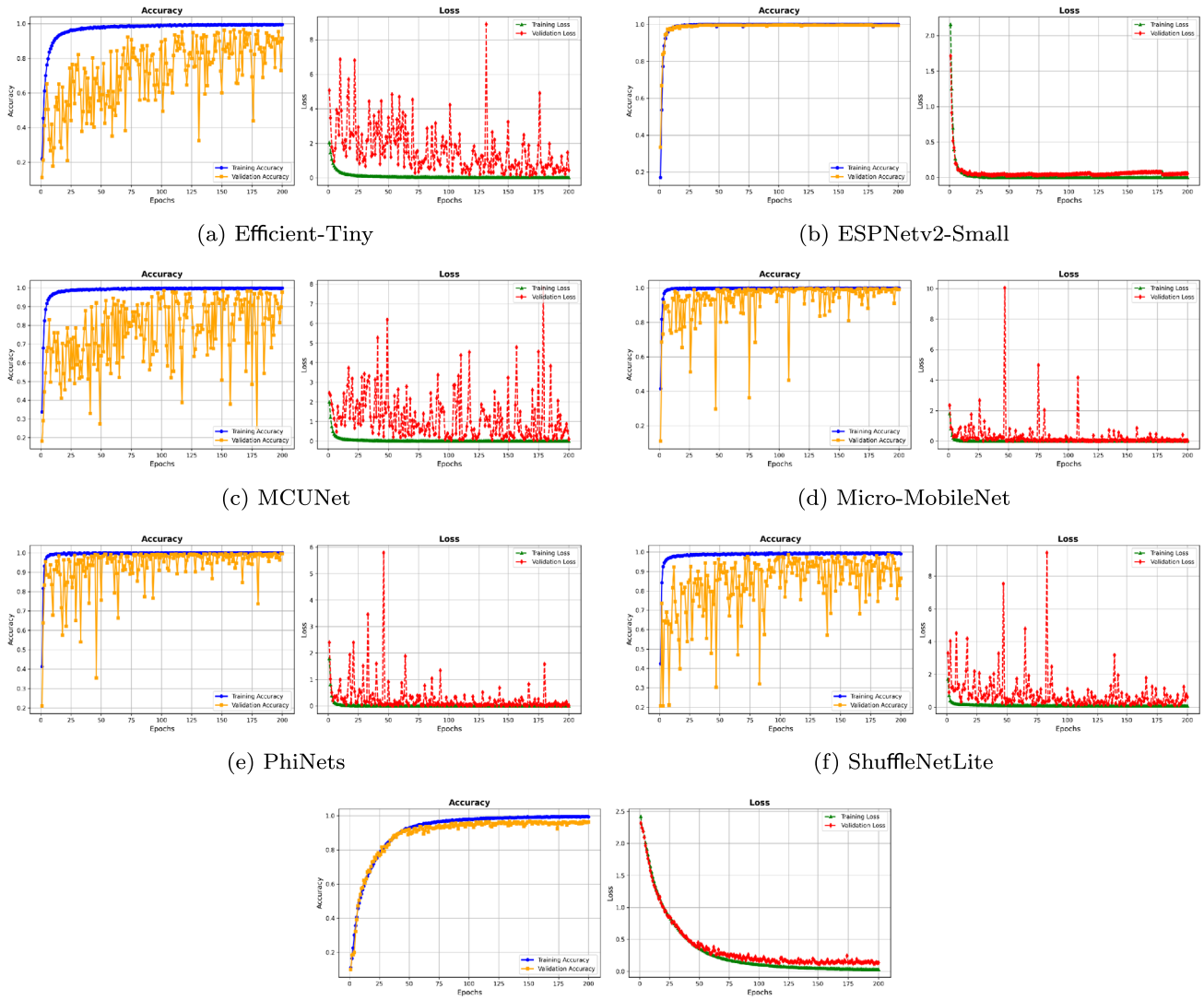
4 Results and discussion

This section presents the evaluation of seven lightweight deep learning architectures for real-time driver state detection. The analysis emphasizes training stability,

generalization performance, and embedded deployment feasibility, assessed through standard classification metrics (accuracy, precision, recall, and F1-score) and computational indicators, such as model size, inference speed, and memory utilization.

4.1 Training and validation performance

An examination of the training and validation accuracy and loss curves across all evaluated architectures (Fig. 2) reveals distinct convergence behaviors and generalization dynamics. Most models achieved a rapid improvement in accuracy during the initial training epochs, followed by gradual stabilization of both accuracy and loss. However, considerable differences were observed in the smoothness and consistency of the validation trajectories. Architectures exhibiting irregular, zigzag-shaped validation curves—most notably Efficient-Tiny and ShuffleNetLite—demonstrated fluctuating generalization performance, suggesting sensitivity to mini-batch variations and possible overfitting. Such oscillatory validation losses indicate that these networks occasionally overadapted to specific training subsets, leading to unstable performance on unseen samples. Efficient-Tiny attained 94.2% training and 91.47% validation accuracy, but its validation loss showed persistent volatility, reflecting an incomplete convergence of its feature representations. In contrast, ESPNetv2-Small exhibited a remarkably stable and monotonic training process, reaching 100% training and 99.48% validation accuracy with minimal loss values. While the model achieved perfect training accuracy, the slight divergence



(a) Efficient-Tiny

(b) ESPNetv2-Small

(c) MCUNet

(d) Micro-MobileNet

(e) PhiNets

(f) ShuffleNetLite

(g) SqueezeNetv11

Fig. 2 Training and validation accuracy/loss curves for the evaluated TinyDL models

between training and validation curves suggests potential mild overfitting, indicating that the model may be capturing dataset-specific patterns. This observation is corroborated by the confusion matrices and test-phase metrics, which offer a more robust evaluation of the model's generalization performance. MCUNet also achieved consistent convergence, obtaining 99.8% training and 97.44% validation accuracy, thereby validating the efficiency of its neural architecture search-based design in balancing compactness and discriminative performance. Micro-MobileNet and PhiNets produced nearly saturated accuracies (99.9–100%) accompanied by very low validation losses (0.02–0.03), confirming the efficacy of depthwise separable and mixed-kernel convolution strategies for preserving representational richness under constrained computational budgets. By contrast, ShuffleNetLite converged more slowly and displayed pronounced fluctuations in validation loss, indicating limited stability and suboptimal adaptation to complex posture variations. SqueezeNetv11 achieved 99.7% training and 96.27% validation accuracy, reflecting an appropriate compromise between aggressive compression and discriminative depth.

Figure 3 presents the relationship between model size and validation accuracy for the evaluated lightweight architectures. The scatter plot reveals that increased model capacity does not necessarily translate to improved predictive performance. Notably, several compact models demonstrate exceptional generalization despite their minimal memory footprint. ESPNetv2-Small (222 KB) and PhiNets (150.9 KB) achieve the highest validation accuracies (99.48% and 99.50%, respectively), underscoring the strength of well-crafted multi-scale and mixed-kernel designs in retaining discriminative capability under stringent resource constraints. Similarly, MCUNet—the smallest architecture assessed at just 34.5 KB—delivers a robust 97.44% validation accuracy, highlighting the benefits of neural architecture search (NAS) in optimizing models for ultra-constrained environments. By contrast, ShuffleNetLite, despite being the largest model (1.75 MB), attains the lowest

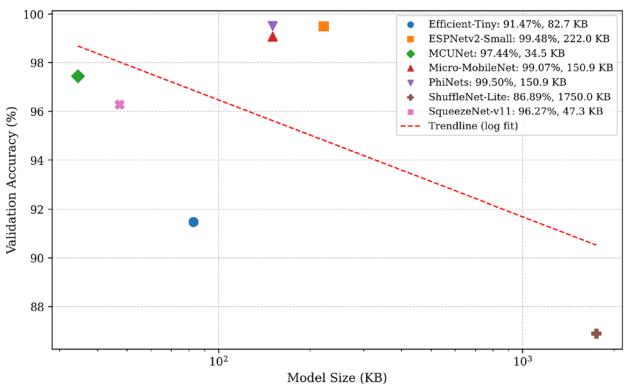


Fig. 3 The relationship between model size and validation accuracy

accuracy (86.89%), suggesting that architectural efficiency plays a more critical role than parameter count in determining generalization. The log-scale trendline further supports this view, revealing a slight inverse correlation between model size and validation accuracy, and reinforcing the notion that compact TinyDL architectures, when thoughtfully designed, can outperform their larger counterparts.

The comparative results demonstrate that ESPNetv2-Small and PhiNets consistently delivered the highest predictive accuracy, convergence stability, and generalization capability. Efficient-Tiny and SqueezeNetv11 achieved satisfactory accuracy with exceptionally small parameter footprints, reaffirming their viability for applications constrained by memory or energy limitations. Although minor fluctuations were observed in their validation curves and moderate confusion occurred in less frequent behavior classes. ShuffleNetLite, in contrast, exhibited the most irregular validation trend and higher off-diagonal confusion, reflecting a tendency toward performance instability. Nevertheless, its superior inference speed and minimal computational cost make it an attractive option for latency-sensitive environments where rapid decision-making outweighs marginal accuracy loss.

4.2 Test performance

The test-phase analysis provides a detailed assessment of the generalization ability and computational efficiency of the evaluated TinyDL architectures. The radar charts in Fig. 4 illustrate the comparative behavior of the seven models across the four main evaluation metrics: accuracy, precision, recall, and F1-score. The results reveal a distinct separation between the top-performing architectures—ESPNetv2-Small and PhiNets—and the remaining models. Both achieved nearly saturated values across all metrics, confirming their capacity to deliver balanced classification performance under resource-constrained conditions. These visual observations are supported by the numerical results in Table 1, where PhiNets achieved the highest test accuracy (99.66%) together with precision, recall, and F1-scores around 99.5%. ESPNetv2-Small followed closely with 99.50% accuracy and consistent performance across the remaining metrics (99.44–99.45%). Micro-MobileNet (98.99%) and MCUNet (97.85%) also demonstrated strong generalization, offering a favorable compromise between predictive performance and model compactness. Efficient-Tiny and SqueezeNetv11, while attaining moderate accuracies (91.64% and 95.81%, respectively), remained advantageous for devices with severe memory and energy constraints due to their extremely low parameter counts. ShuffleNetLite, in contrast, recorded the lowest test accuracy (86.06%) and F1-score (87.21%), indicating that its minimal architecture trades off precision and recall for higher inference speed (22.27 FPS).

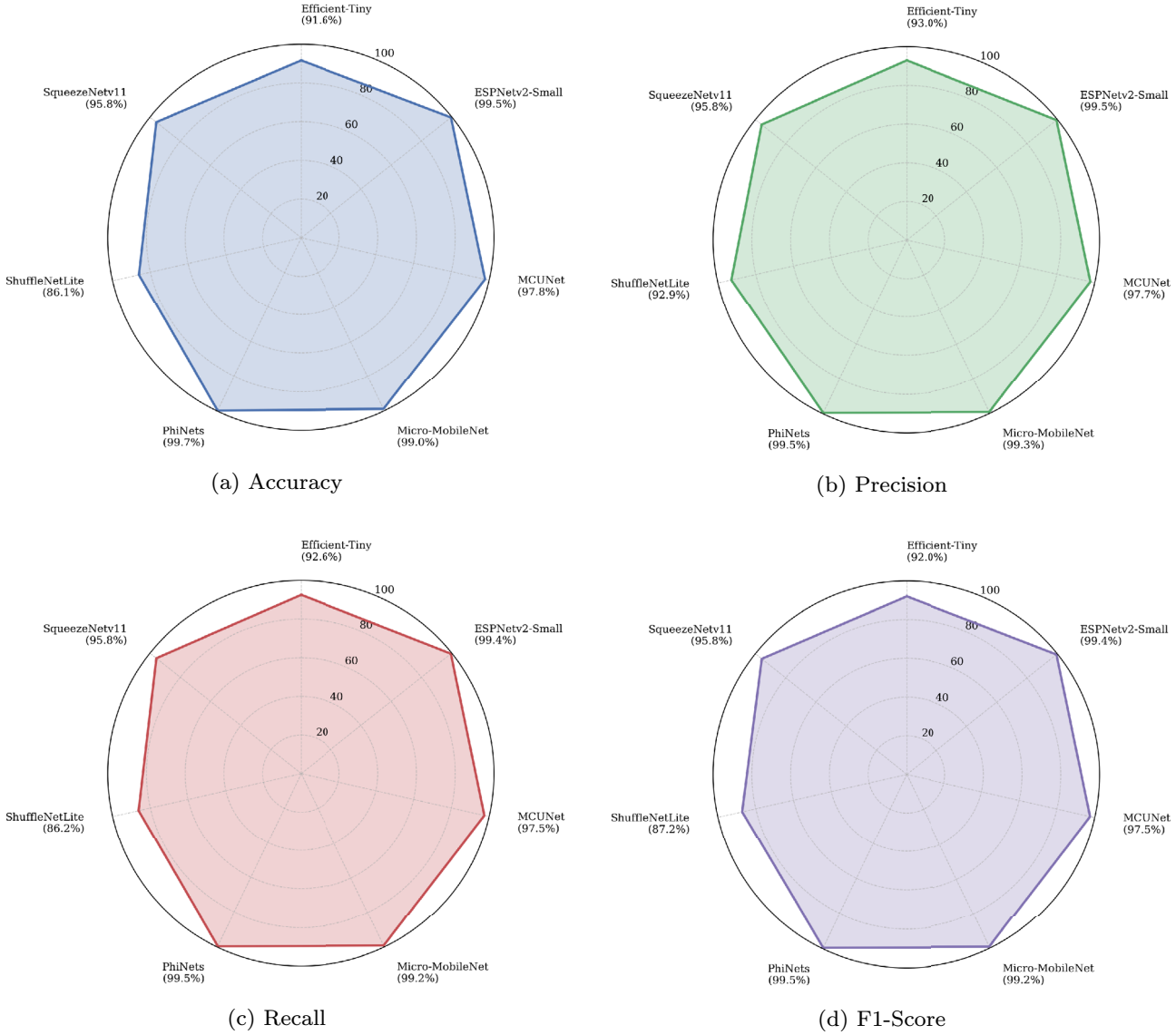


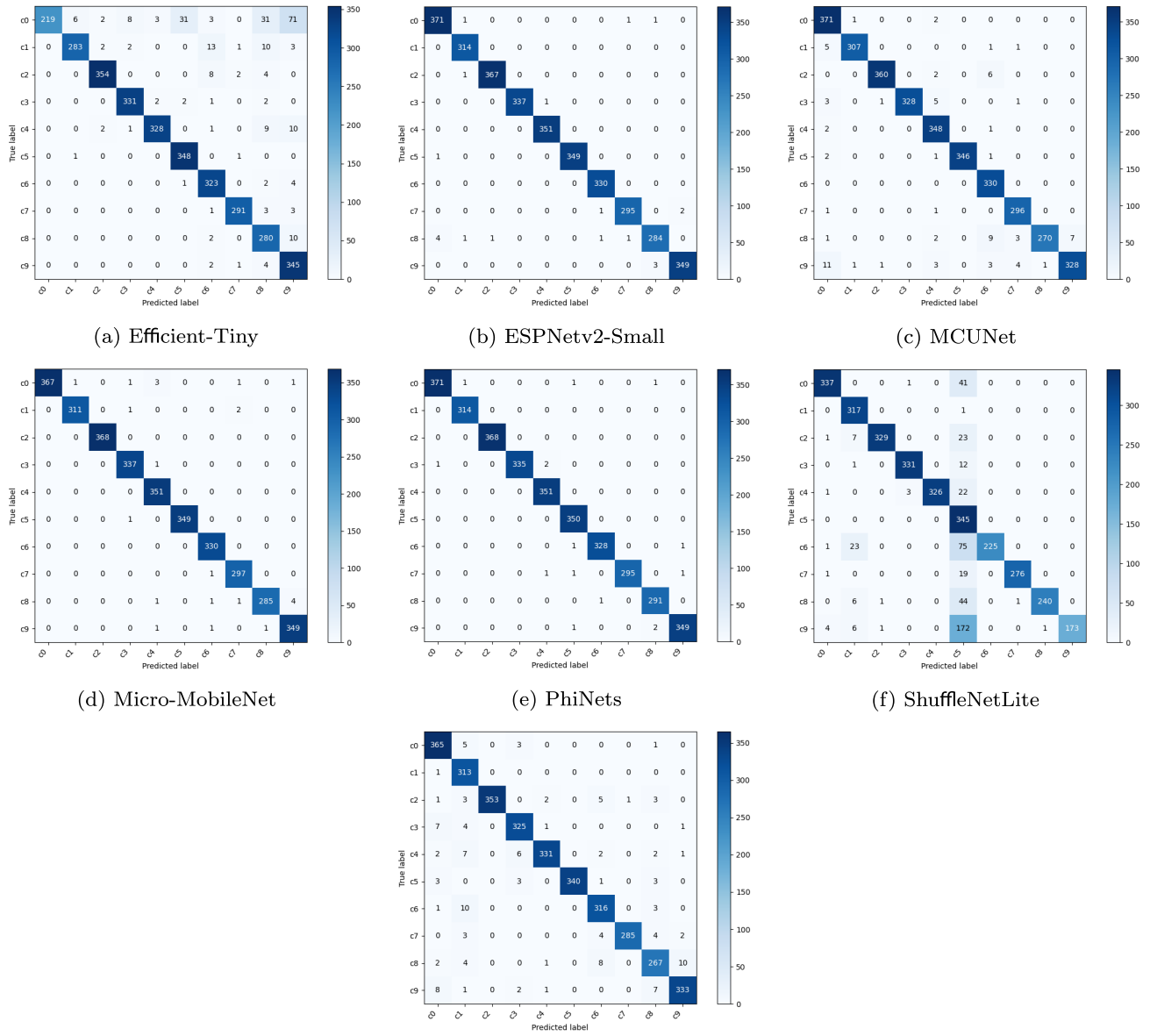
Fig. 4 Radar chart visualization of the main performance metrics for the evaluated lightweight models: (a) accuracy, (b) precision, (c) recall, and (d) F1-Score

Table 1 Performance comparison of lightweight models in terms of test metrics, computational efficiency, and memory footprint

Model	Accuracy (%)	Loss	Precision (%)	Recall (%)	F1-score (%)	RAM (MB)	Speed (FPS)	Size (MB)
Efficient-Tiny	91.64	0.4243	93.02	92.56	92.01	0.08	26.81	0.43
ESPNetv2-Small	99.50	0.0482	99.45	99.44	99.44	0.22	40.49	0.74
MCUNet	97.85	0.0951	97.69	97.48	97.54	0.03	52.72	0.20
Micro-MobileNet	98.99	0.0300	99.26	99.23	99.24	0.15	37.71	0.60
PhiNets	99.66	0.0175	99.50	99.49	99.49	0.15	36.14	0.60
ShuffleNetLite	86.06	0.6538	92.93	86.23	87.21	1.75	22.27	5.46
SqueezeNetv11	95.81	0.1399	95.85	95.76	95.76	0.05	28.60	0.33

The confusion matrices shown in Fig. 5 provide additional insight into class-level behavior. ESPNetv2-Small, PhiNets, and Micro-MobileNet exhibit clear diagonal dominance with minimal off-diagonal dispersion, indicating that their high macro-level performance results from consistently accurate classification across all driver behavior categories rather than bias toward frequent classes. MCUNet maintains similar characteristics, with misclassifications primarily appearing among visually related activities, such as texting and phone usage. Efficient-Tiny and SqueezeNetv11 display moderate confusion in less represented or ambiguous behaviors, reflecting partial limitations in capturing subtle pose variations. ShuffleNetLite shows the most scattered confusion pattern, consistent with its reduced accuracy and recall metrics.

Among the evaluated architectures, the analysis indicates a clear hierarchy of performance and suitability for embedded deployment. ESPNetv2-Small and PhiNets deliver the most reliable and generalizable results among the tested models, combining high accuracy, stable convergence, and compact design. Micro-MobileNet and MCUNet achieve an effective balance between discriminative performance and computational cost, which is advantageous for real-time embedded deployment. Efficient-Tiny and SqueezeNetv11



(a) Efficient-Tiny (b) ESPNetv2-Small (c) MCUNet

(d) Micro-MobileNet (e) PhiNets (f) ShuffleNetLite

(g) SqueezeNetv11

Fig. 5 Test confusion matrices for the evaluated lightweight models

remain suitable for environments dominated by hardware limitations, while ShuffleNetLite is more appropriate for latency-sensitive scenarios that demand rapid inference rather than peak precision. These outcomes confirm the practicality of deploying TinyDL models for real-time driver state detection without compromising predictive reliability or system efficiency.

4.3 Discussion

The comparative analysis of seven lightweight deep learning architectures confirms that modern TinyDL models can achieve competitive accuracy levels while maintaining excellent computational efficiency and deployability in embedded driver monitoring applications. The findings obtained from the training, validation, and testing phases underscore the decisive influence of architectural design on learning stability, convergence behavior, and generalization capability.

Among all evaluated networks, ESPNetv2-Small and SqueezeNetv11 consistently exhibited superior stability and predictive performance. The use of dilated spatial pyramid modules in ESPNetv2-Small and fire modules in SqueezeNetv11 enabled efficient multi-scale feature extraction while minimizing computational overhead. These architectures achieved test accuracies of 99.50% and 95.81%, respectively, with compact model sizes of 0.74 MB and 0.33 MB, peak RAM consumption of 0.22 MB and 0.05 MB, and real-time inference speeds of 40.49 FPS and 28.60 FPS. Their smooth convergence profiles and low loss values indicate effective optimization and robust generalization, making them highly suitable for real-time embedded and TinyML deployment. In contrast, Efficient-Tiny, despite its smaller footprint (0.43 MB) and moderate inference speed (26.81 FPS), exhibited unstable validation behavior and overfitting, yielding a lower test accuracy of 91.64%. These observations highlight the inherent limitation of extreme compression, which can reduce representational capacity and impair feature discrimination in complex posture recognition tasks.

From a deployment perspective, model size and peak RAM usage represent key indicators of feasibility for ultra-constrained microcontrollers such as the Arduino Nano BLE Sense, which provides approximately 1 MB of Flash memory and 256 KB of SRAM. Both ESPNetv2-Small and SqueezeNetv11 fall well within these limits, confirming their direct deployability without architectural modification. Their compact memory requirements and high inference speeds make them ideal candidates for on-device inference in real-world driver monitoring scenarios. In contrast, larger architectures such as ShuffleNetLite (1.75 MB, 1.75 MB RAM) exceed the microcontroller's available memory and thus require quantization or structured pruning to enable feasible deployment. Similarly, models including MCUNet and Micro-MobileNet, while already efficient, would further benefit from INT8 quantization and mixed-precision inference to reduce latency and energy consumption without compromising accuracy.

Overall, these results demonstrate that ESPNetv2-Small and SqueezeNetv11 offer the most favorable balance between accuracy, memory efficiency, and inference speed, rendering them optimal for direct deployment in low-power embedded environments. Conversely, architectures, such as Efficient-Tiny and ShuffleNetLite, require additional optimization to improve their generalization and resource utilization. The findings emphasize that the design of lightweight architectures extends beyond parameter reduction and must preserve a balance between network depth, width, and receptive-field resolution. By combining multi-scale feature aggregation, depthwise separability, and efficient scaling, well-calibrated architectures, such as ESPNetv2-Small and SqueezeNetv11, demonstrate strong potential for deployment in next-generation TinyML-based intelligent transportation systems, including driver monitoring, drowsiness detection, and in-cabin behavior analysis.

In comparison with recent state-of-the-art studies summarized in Table 2—all of which report results based on the same StateFarm dataset—the present work demonstrates competitive, and in some cases superior, accuracy while relying on significantly smaller and more resource-efficient architectures. Prior approaches, including MobileNetV3 and ShuffleNetV2 [4, 15], as well as hybrid NAS-based models such as CCNET [25] and CoViT [22], have achieved high recognition performance, with reported accuracies ranging from approximately 97 to 99.9%. Other optimized CNN architectures, including ATD2Net [26] and Si-CA MobileNet [28], have also demonstrated strong results. However, these models typically exhibit higher architectural complexity and parameter counts.

By contrast, the lightweight architectures evaluated in this study—particularly ESPNetv2-Small and SqueezeNetv11—attain comparable levels of performance while maintaining significantly reduced parameter sizes. Specifically, ESPNetv2-Small achieves 99.50% accuracy, and SqueezeNetv11 attains 95.81%, underscoring the effectiveness of compact design in achieving high accuracy under tight resource constraints.

5 Challenges, limitations, and future work

Despite the promising results, several challenges remain for fully optimized TinyDL deployment in real-time driver state detection on embedded and low-power systems. Key limitations involve hardware constraints, activation memory

Table 2 Comparison of the proposed models with state-of-the-art lightweight architectures using the StateFarm dataset

Study (year)	Model	Parameters (M)	Accuracy (%)	FPS	Type
Li et al. [4] (2022)	MobileNetV3 (Multi-Teacher KD)	0.35	+5.63 gain	42	Lightweight
Li et al. [4] (2022)	ShuffleNetV2 (Multi-Teacher KD)	0.35	+8.41 gain	25	Lightweight
Nguyen et al. [15] (2022)	MobileNetV3-small + CBAM	0.43	99.95	243	Lightweight
Huang et al. [16] (2023)	PSDNet	1.33	91.70	52	Lightweight
Li et al. [22] (2024)	CoViT (CNN + ViT)	1.24	97.89	159	Lightweight
Sun & Zhang [25] (2025)	CCNET (NAS + CA)	1.70	99.8–⁻99.9	54	Lightweight
Lv et al. [28] (2025)	Si-CA MobileNet	3.17	99.80	83	Lightweight
Gu et al. [26] (2025)	ATD2Net	3.21	96.40	29	Lightweight
Our Study (2025)	PhiNets	0.038	99.66	36	Tiny
	ESPNetv2-Small	0.057	99.50	40	Tiny
	Micro-MobileNet	0.039	98.99	38	Tiny
	MCUNet	0.009	97.85	53	Tiny
	Efficient-Tiny	0.021	91.64	27	Tiny
	SqueezeNetv11	0.012	95.81	29	Tiny
	ShuffleNetLite	0.458	86.06	22	Lightweight

bottlenecks, model compression trade-offs, and the absence of post-training optimization techniques.

Hardware restrictions—including limited Flash and SRAM capacity—pose a fundamental constraint to the feasibility of deploying larger architectures, as runtime memory usage often exceeds the available resources on microcontrollers, thus requiring hardware-aware co-design. Although models, such as ESPNetv2-Small, PhiNets, and MCUNet, are designed for low-memory operation, activation memory can still surpass the model's static footprint, especially in architectures utilizing multi-scale or dilated convolutions [37].

Ultra-compact models, such as Efficient-Tiny and SqueezeNetv11, also experience noticeable accuracy degradation, primarily due to aggressive pruning and kernel reduction strategies [38]. These results highlight the importance of adaptive compression methods that preserve critical feature representations while minimizing redundancy.

Moreover, this study did not implement post-training deployment-level optimization methods, such as quantization, structured pruning, or mixed-precision inference, which are known to significantly reduce model size, latency, and power consumption. Future research will integrate these optimization strategies [39] and conduct extensive benchmarking on representative microcontroller platforms, including Arduino Nano BLE Sense, STM32 Nucleo, and ESP32-CAM [40], evaluating real-time inference latency, throughput, and energy efficiency.

Additionally, the current framework will be extended to support multimodal driver monitoring by integrating complementary visual, behavioral, and physiological modalities, such as gaze direction, hand gestures, and voice activity [6]. Finally, future efforts will focus on full system integration and in-vehicle validation, emphasizing energy-aware scheduling strategies and edge–cloud collaboration [41] to ensure scalability, privacy preservation, and low-latency operation in real-world deployments.

6 Conclusions

This study evaluated seven lightweight deep learning architectures—Efficient-Tiny, ESPNetv2-Small, MCUNet, Micro-MobileNet, PhiNets, ShuffleNetLite, and SqueezeNetv11—for real-time driver state detection on embedded systems. PhiNets and ESPNetv2-Small achieved the best test performance, with accuracies of 99.66% and 99.50%, F1-scores exceeding 99.4%, and compact model sizes below 1 MB. Micro-MobileNet and MCUNet also demonstrated strong results, achieving 98.99% and 97.85% accuracy, with minimal memory usage (0.15 MB and 0.03 MB) and real-time inference speeds (37.71 and 52.72 FPS). Efficient-Tiny and SqueezeNetv11 maintained acceptable accuracy (91.64% and 95.81%), while ShuffleNetLite provided the fastest inference (22.27 FPS) at the cost of lower accuracy (86.06%).

These results confirm the practical feasibility of TinyDL models for embedded driver monitoring. However, challenges persist due to memory constraints and the absence of post-training optimizations, such as INT8 quantization or structured pruning. Future work will focus on hardware-specific model compression and the integration of multimodal input signals—including facial expressions, hand gestures, and voice activity—to improve robustness under real-world driving conditions.

Author Contributions S. E led the research, performing the core experimental design, model benchmarking, analysis, and manuscript writing. A. E and Q. Z contributed to methodology and validation. N. A assisted with data curation and review. S. S provided the supervision,

Data Availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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